1a) Velocity profile:

$$V = \frac{(\Delta P + \rho g \Delta z) D^2}{16 \mu l} \left[1 - \left(\frac{2r}{D} \right)^2 \right]$$

Centreline velocity is when $r = 0$,

$$V_c = \frac{(\Delta P + \rho g \Delta z) D^2}{16 \mu l} \left[1 - \left(\frac{2(0)}{D} \right)^2 \right]$$

$$= \frac{(\Delta P + \rho g \Delta z) D^2}{16 \mu l}$$

$$\therefore V = V_c \left[1 - \left(\frac{2r}{D} \right)^2 \right]$$

Volume flow rate:

$$Q = \int_A V dA$$

$$= \int_0^{\frac{D}{2}} V_c \left[1 - \left(\frac{2r}{D} \right)^2 \right] 2\pi r dr$$

$$= 2\pi V_c \int_0^{\frac{D}{2}} r - \frac{4r^3}{D^2} dr$$

$$= 2\pi V_c \left[\frac{r^2}{2} - \frac{\cancel{4} r^4}{4D^2} \right]_0^{\frac{D}{2}}$$

$$= 2\pi V_c \left[\frac{D^2}{8} - \frac{D^{\cancel{4}2}}{16 \cancel{D^2}} \right]$$

$$= \frac{\pi V_c D^2}{8}$$

1a) $V_{avg} = \frac{Q}{A}$

$$= \frac{\cancel{\pi} V_c \cancel{D^2}}{\cancel{8}\,2} \div \frac{\cancel{\pi} \cancel{D^2}}{\cancel{4}}$$

$$= \frac{1}{2} V_c \text{ (shown)}$$

b) $D = 0.1$, $V_c = 3$, $\rho = 1260$, $\mu = 0.9$

$$Re = \frac{\rho V_{avg} D}{\mu} = \frac{\rho \frac{V_c}{2} D}{\mu}$$

$$= \frac{1260\left(\frac{3}{2}\right)(0.1)}{0.9}$$

$$= 210$$

$$V_c = \frac{(\Delta P + \rho g \Delta z) D^2}{16 \mu l}$$

$$0.3 = \frac{(\Delta P + \cancel{\rho g (0)})(0.1)^2}{16(0.9) l}$$

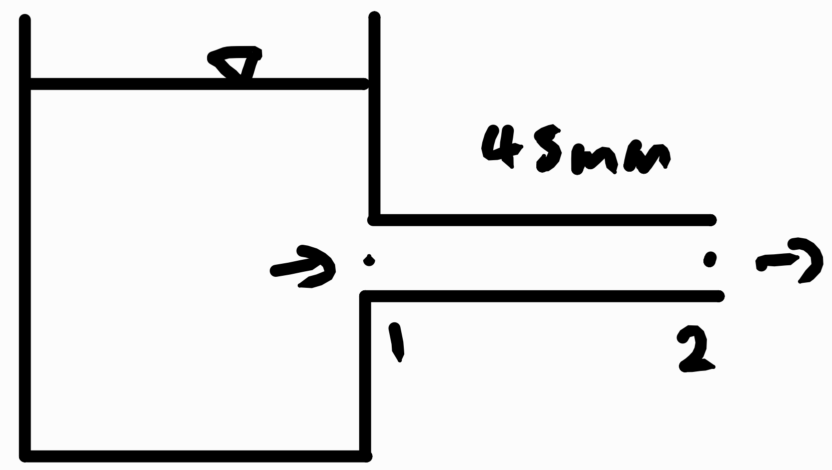
$$0.3 = \frac{0.01 \Delta P}{14.4 l}$$

$$\frac{\Delta P}{l} = 432 \text{ Pa m}^{-1}$$

The fluid flows from high to low pressure, hence the pressure gradient is negative.

$$\therefore \frac{\Delta P}{l} = -4.32 \text{ kPa m}^{-1}$$

2)



$S = 0.95$

$\rho = 950 \text{ kgm}^{-3}$

$$V = \frac{Q}{A} = \frac{0.12 \times 10^{-3} \times 10^{-3}}{\pi \frac{(0.75 \times 10^{-3})^2}{4}}$$

$$= 0.2716244362$$

$$\approx 0.272 \text{ ms}^{-1}$$

Using Bernoulli's equation:

$$\frac{P_1}{\rho g} + \cancel{\frac{V_1^2}{2g}} + \cancel{z_1} = \cancel{\frac{P_2}{\rho g}} + \cancel{\frac{V_2^2}{2g}} + \cancel{z_2} + h_L$$

$$\frac{P_1}{\rho g} = h_L$$

$$\frac{P_1}{\rho \cancel{g}} = f \frac{l}{D} \frac{V_{avg}^2}{2\cancel{g}}$$

$$f = \frac{2 P_1 D}{\rho l V_{avg}^2}$$

$$= \frac{2(1.31\cancel{\times 10^{3}})(0.75\cancel{\times 10^{-3}})}{950 \times 45 \times 10^{-3} (0.272)^2}$$

$$= 0.6230010231 \approx 0.623$$

2) Assuming laminar flow:

$$f = \frac{64}{Re}$$

$$f = \frac{64\mu}{\rho V_{avg} D}$$

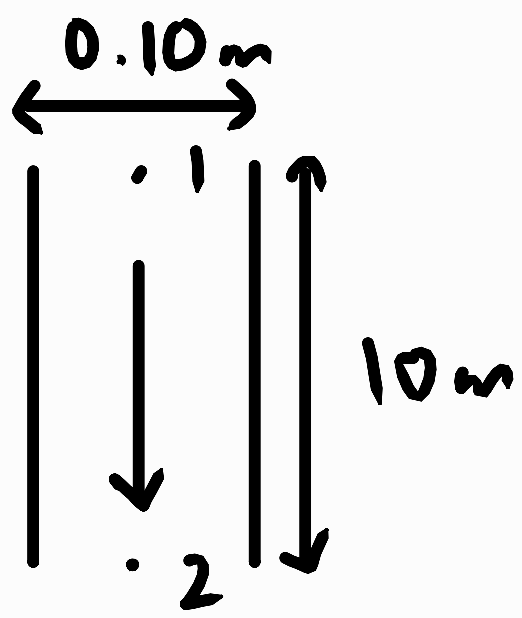
$$\mu = \frac{f \rho V_{avg} D}{64}$$

$$= \frac{0.623(950)(0.272)(0.75\times10^{-3})}{64}$$

$$= 0.00188392 \text{ kg m}^{-1}\text{s}^{-1}$$

$$\approx 0.00188 \text{ kg m}^{-1}\text{s}^{-1}$$

3) $\rho = 1000\, kg\, m^{-3}$ $\quad \mu = 0.30\, Ns\, m^{-2}$



Maximum Re for laminar flow:

$$Re = 2100$$

$$Re = \frac{\rho V D}{\mu}$$

$$2100 = \frac{1000 v (0.10)}{0.3}$$

$$V_{avg} = 6.3\, ms^{-1}$$

$$V_{avg} = \frac{D^2(\Delta P - \rho g \Delta z)}{32 \mu L}$$

$$\Delta P = \frac{32 \mu L V_{avg}}{D^2} + \rho g \Delta z$$

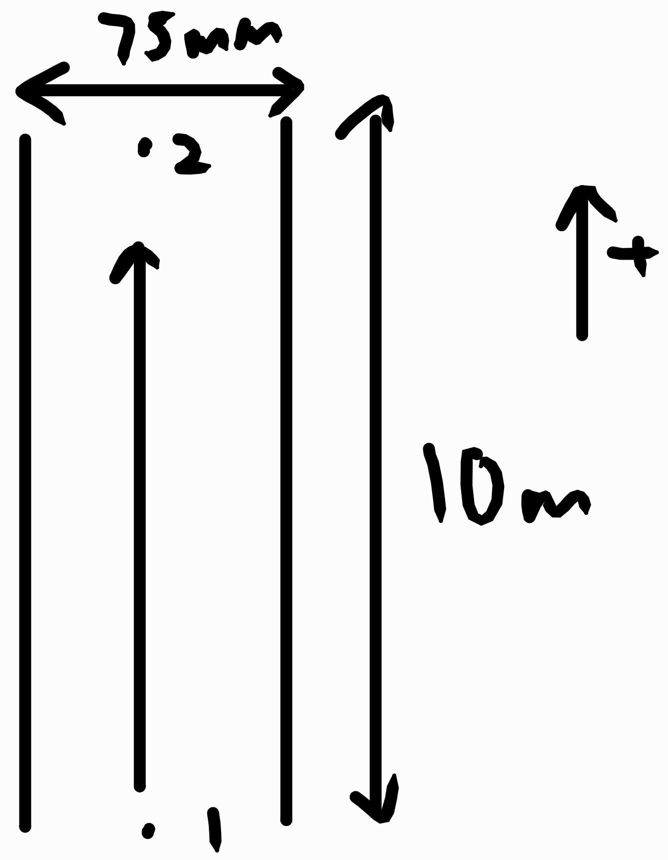
$$P_1 - P_2 = \frac{32(0.1)(10)(6.3)}{0.1^2} + 1000(9.81)(-10)$$

$$P_1 - 0 = -37620\, Pa$$

$$P_1 = -37620\, Pa$$

$$= -37.62\, kPa$$

4)



$V_c = 1.0\,ms^{-1}$ $\quad \mu = 1.5\,Ns\,m^{-2}$

$V_{avg} = 0.5\,ms^{-1}$ $\quad \rho = 1260\,kg\,m^{-3}$

$$Re = \frac{\rho V D}{\mu}$$

$$= \frac{1260\left(\frac{1}{2}\right)(75\times10^{-3})}{1.5}$$

$$= 31.5 \leq 2100$$

The flow is laminar.

$$V_{avg} = \frac{D^2(\Delta P - \rho g \Delta z)}{32\mu L}$$

$$\Delta P = \frac{32\mu L V_{avg}}{D^2} + \rho g \Delta z$$

$$= \frac{32(1.5)(10)(0.5)}{(75\times10^{-3})^2} + 1260(9.81)(10)$$

$$= 166272.6667\ Pa$$

$$\approx 166\ kPa$$

$$h_L = f\,\frac{L}{D}\,\frac{V_{avg}^2}{2g}$$

$$= \frac{64}{31.5}\left(\frac{10}{75\times10^{-3}}\right)\left(\frac{0.5^2}{2\times9.81}\right)$$

$$= 3.451828121$$

$$\approx 3.45\ m$$